

Volume I · Chapter 4 — Core Machine Learning for Medicine · Labs

Source: split from med-tech-curriculum-V1.md (V1). From this split onward this chapter is maintained **independently** here. See Volume-I-Chapter-4-context.md for the AI interaction log.

Prerequisites: Ch 1 (NumPy/Pandas), Ch 2 (clinical data).

Learning outcomes — the student can:

- Frame a clinical problem as **supervised or unsupervised** ML and build/train/evaluate models with **scikit-learn**.
- Apply a rigorous **workflow** (train/val/test, cross-validation) and avoid **data leakage**.
- Choose and **interpret clinically appropriate metrics** (sensitivity/specificity, PPV/NPV, ROC-AUC, PR-AUC, calibration).
- Use **unsupervised** methods (clustering, PCA) for **patient stratification**, and recognize imbalance/bias issues.

Labs (hands-on) — 16 h:

#	Lab	h
4a	Build a readmission classifier (logistic regression) with a proper train/test split	3
4b	Random forest / gradient boosting on the same task; compare; feature importance	3
4c	Evaluation deep-dive: ROC/PR curves, calibration, clinical threshold selection	3
4d	Cross-validation & hyperparameter tuning; leakage checks	3
4e	Unsupervised patient stratification: clustering + PCA visualization	4

Datasets/tools: synthetic structured EHR; scikit-learn, Pandas, Matplotlib.

Assessment: modeling project — build + evaluate a classifier with written interpretation (50%); evaluation/metrics lab rubric (30%); quiz (20%).

Key decisions: metric choice for the clinical question; **sensitivity vs. specificity** trade-off; complexity vs. **interpretability**; handling imbalance; avoiding leakage.

References: plan §13 → *Programming & data science*; *Clinical datasets*.

Hours: Theory **14** + Lab **16** = **30**. (*Feeds plan Lab 2 — Simple Clinical Classifier.*)